

TONY HAN

☎ (403) · 836 · 1476 ✉ hhaiyang.ren@gmail.com 🌐 github.com/1tonyhan 🔗 LinkedIn.com/in/1tonyhan

EDUCATION

University of British Columbia

Bachelor's in Applied Science (BASc), Engineering Physics.

Vancouver, BC

Expected graduation 2029. Dean's list (2024-2025).

SKILLS

Programming:	Python, LaTeX, C, Java.
Development Tools:	Altium Designer, Onshape, Git, Microsoft Excel, Adobe Photoshop, Canva.
Hardware & Design:	PCB Design, Arduino, ESP32, Oscilloscope, Analog Discovery 3, Soldering, 3D Printing, Lathe, Mill.

TECHNICAL EXPERIENCE

UBC Orbit Satellite Design Team

Admin Team Lead

January 2025 - Present

- Lead a multidisciplinary team of 5 to handle administrative logistics for an 80+ member design team to support satellite development.
- Developed a Python automation tool to parse purchase records and automatically generate reimbursement forms, eliminating manual data entry and significantly reducing processing time.
- Established documentation and project-tracking workflows that improved coordination between technical and administrative teams, supporting efficient project execution.

On-Board Computer (OBC) subteam member

January 2026 - Present

- Designed communication interface PCBs in Altium Designer to enable communication between the OBC and the satellite reaction wheels system, optimizing for signal integrity and board manufacturability.
- Developed harness routing models in Onshape to evaluate cable management and identify assembly constraints prior to fabrication and engineering model integration.
- Identified and resolved inter-subsystem integration constraints, ensuring electrical and mechanical compatibility between onboard computer PCBs and attitude control hardware.

TECHNICAL PROJECTS

Autonomous Competition Robot

June 2026 - present

- Designed, manufactured, and validated a servo-actuated multifunctional claw capable of performing multiple competition objectives, using CAD modeling and prototyping equipment to support iterative improvements.
- Optimized model geometry for additive manufacturing and part integration/assembly, reducing filament waste and print time by 25% while maintaining functional performance.
- Developed mechanical components in Onshape, translating design concepts into manufacturable, 3D-printable parts for robot subsystems and assemblies.
- Created and executed validation procedures to evaluate design performance, enabling rapid iteration and data-driven refinement throughout the development cycle.

Motor Driver control circuit

September 2025 - December 2025

- Designed, implemented and debugged a closed-loop DC motor control system incorporating speed feedback and an R-2R DAC-based control interface.
- Utilized Analog Discovery 3 and oscilloscope to diagnose signal integrity issues and validate circuit performance.
- Developed Jupyter Notebook and Python-based documentation with annotated images and test procedures to streamline troubleshooting and system validation.